

Advanced Statistical Learning

Model Assessment & Selection

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Outline

The Bias-Variance Decomposition

Optimism of Training Error Rate

Estimates of Err_{in}

- C_p Statistic

- AIC

- Effective Number of Parameters

- Bayesian Approach & BIC

Cross Validation

Bootstrap Method

Regression - A Review

Estimate target variable Y from a vector of inputs X .

A prediction model $\hat{f}(X)$ estimated from training data

$$T = \{(x_1, y_1), \dots, (x_n, y_n)\}$$

The loss function $L(Y, \hat{f}(X))$ measures the error / loss between Y and $\hat{f}(X)$

Common choices for loss function

$$L(Y, \hat{f}(X)) = \begin{cases} (Y - \hat{f}(X))^2, & \text{squared error} \\ |Y - \hat{f}(X)|, & \text{absolute error} \end{cases}$$

Test Error

Test error, also called generalization error

$$Err_T = E[L(Y, \hat{f}(X)) \mid T]$$

- ▶ Prediction error over an **independent** test sample
- ▶ X and Y are drawn randomly from $p(X, Y)$
- ▶ The training set T is fixed

Expected Prediction Error

Expected prediction error, also called expected test error

$$Err = E[L(Y, \hat{f}(X))] = E[Err_T]$$

- ▶ Average over everything that is random
- ▶ Including the randomness in the training set that produced $\hat{f}$

We would be interested in estimating Err_T , but in most cases it is easier to estimate Err

Training Error

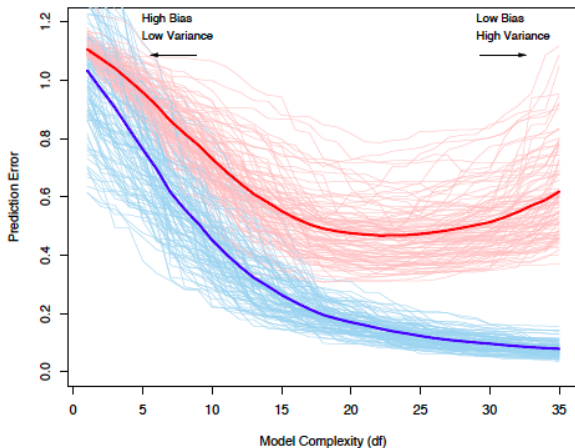
Training error

$$\overline{err} = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(x_i))$$

As model complexity increases

- ▶ $\overline{err} \rightarrow 0$
- ▶ but Err_T increases, tendency to overfit
- ▶ $\overline{err}$ is not a good estimate of Err_T or Err

Errors as Model Complexity Increases



1. Light Blue Curve: training error $\overline{err}$
2. Solid blue curve: expected training error $E[\overline{err}]$
3. Light red curve: conditional test error Err_T
4. Solid red curve: expected test error Err

Classification - A Review

Estimate categorical response $G \in \{1, \dots, K\}$ from a vector of inputs X

Typically model $p_k(X) = \Pr(G = k \mid X)$ and define
 $\hat{G}(X) = \operatorname{argmax}_k p_k(X)$

Common choices of loss function

► 0-1 loss

$$L(G, \hat{G}(X)) = \operatorname{Ind}(G \neq \hat{G}(X))$$

► log-likelihood (deviance)

$$L(G, \hat{p}(X)) = -2 \sum_{k=1}^K \operatorname{Ind}(G = k) \log \hat{p}_k(X) = -2 \log \hat{p}_G(X)$$

Ultimate Goal

$$\hat{\alpha} = \operatorname{argmin}_{\alpha} E[L(Y, \hat{f}_{\alpha}(X))]$$

- ▶ $\hat{f}_{\alpha}(x)$ typically has a tuning parameter α controlling its complexity
- ▶ Estimate $E[L(Y, \hat{f}_{\alpha}(X))]$ for different values of α
- ▶ Find the value of α that produces the minimum expected prediction error

Two Separate Goals

Model selection:

- ▶ Estimate the performance of different models in order to choose the best one

Model assessment:

- ▶ Having chosen a final model, estimate its prediction error (generalization error) on new data.

For a Data-rich Situation

Randomly divide the dataset into 3 parts



Common split ratio 50%, 25%, 25%

Model selection

- ▶ Use training set to fit each model
- ▶ Use validation set to estimate Err_T for each model
- ▶ Choose model with lowest Err_T estimate

Model assessment

- ▶ Use the test set - unseen until this stage - to estimate Err_T

For a Data-insufficient Situation

Approximate the validation step either

- ▶ analytically with approaches such as

1. Akaike Information Criterion
2. Bayesian Information Criterion
3. Minimum Description Length
4. Structural Risk Minimization

or

- ▶ with efficient sample re-use

1. cross-validation
2. the bootstrap

Each method also provides estimates of the Err or Err_T of the final chosen model

Bias-Variance Decomposition

The Bias-Variance Decomposition

Assume an additive model

$$Y = f(X) + \epsilon$$

where $E(\epsilon) = 0$ and $Var(\epsilon) = \sigma_\epsilon^2$

The **expected prediction error** of $\hat{f}(X)$ at $X = x_0$

$$Err(x_0) = E((Y - \hat{f}(x_0))^2 \mid X = x_0)$$

can be expressed as

$$Err(x_0) = IrreducibleError + Bias^2 + Variance$$

- ▶ **Irreducible error:** σ_ϵ^2
- ▶ **Bias:** $E[f(x_0) - E[\hat{f}(x_0)]]$
- ▶ **Variance:** $Var[\hat{f}(x_0)] = E[\hat{f}(x_0) - E\hat{f}(x_0)]^2$

K-Nearest-Neighbor Regression Fit

Assuming x_i are fixed,

$$Err(x_0) = \sigma_\epsilon^2 + \left[f(x_0) - \frac{1}{k} \sum_{l=1}^k f(x_{(l)}) \right]^2 + \frac{\sigma_\epsilon^2}{k}$$

- ▶ Complexity of model is inversely related with k
- ▶ As k increases, the variance decreases
- ▶ As k increases, the squared bias increases

Linear Model - least square fit

Now assume a linear model

$$\hat{f}_p(x) = x^\top \hat{\beta}$$

where $\hat{\beta}$ is p -dimensional and fit by least squares, then

$$Err(x_0) = \sigma_\epsilon^2 + [f(x_0) - E(\hat{f}_p(x_0))]^2 + \|h(x_0)\|^2 \sigma_\epsilon^2$$

with $h(x_0) = X(X^\top X)^{-1}x_0$ and $\hat{f}_p(x_0) = x_0^\top (X^\top X)^{-1}X^\top y$

Linear Model - Ridge Fit

Still assume a linear model

$$\hat{f}_{p,\alpha}(x) = x^\top \hat{\beta}_\alpha$$

where $\hat{\beta}_\alpha$ is p -dimensional and fit via ridge regression, then

$$Err(x_0) = \sigma_\epsilon^2 + [f(x_0) - E(\hat{f}_{p,\alpha}(x_0))]^2 + |h_\alpha(x_0)|^2 \sigma_\epsilon^2$$

with $h_\alpha(x_0) = X(X^\top X + \alpha I)^{-1}x_0$ and $\hat{f}_{p,\alpha}(x_0) = x_0^\top (X^\top X + \alpha I)^{-1}X^\top y$

Therefore, this regression fit model has a different bias and variance to the least square fit

Linear Model - Fine Decomposition of Bias

Let β_* denote the parameters of the best-fitting linear approx to f :

$$\beta_* = \operatorname{argmin}_{\beta} E[(f(X) - X^T \beta)^2]$$

Now re-write the averaged squared bias

$$E_{x_0}[(f(x_0) - E[\hat{f}_{\alpha}(x_0)])^2]$$

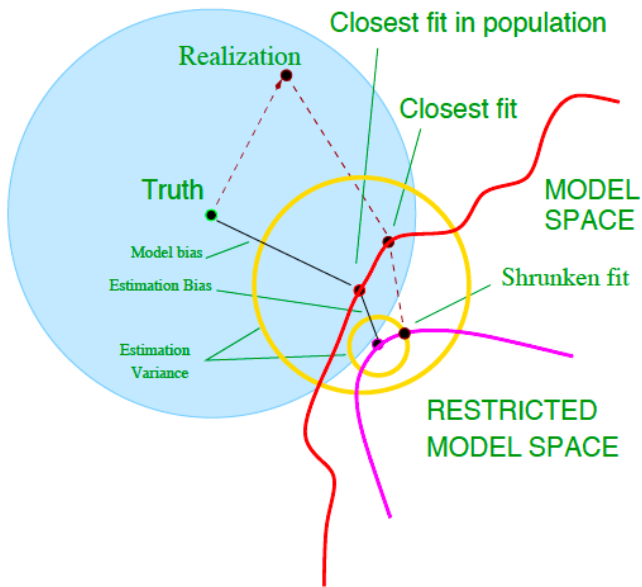
as

$$E_{x_0}[(f(x_0) - x_0^T \beta_*)^2] + E_{x_0}[(x_0^T \beta_* - E[x_0^T \hat{\beta}_{\alpha}])^2]$$

where

- ▶ First term: $\operatorname{Ave}[\text{Model Bias}]^2$
- ▶ Second term: $\operatorname{Ave}[\text{Estimation Bias}]^2$

Estimation bias is zero for least square estimate, and positive for ridge regression estimate



Example 1

Set-up

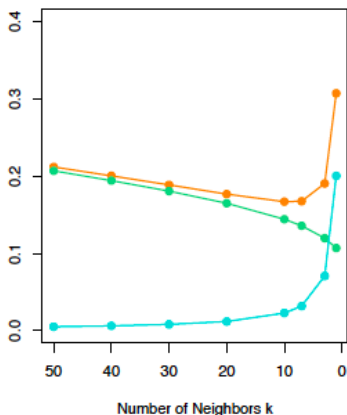
- ▶ $n = 80$ observations and $p = 20$ predictors
- ▶ X is uniformly distributed in $[0, 1]^{20}$ and

$$Y = \begin{cases} 0 & \text{if } X_1 \leq 0.5 \\ 1 & \text{if } X_1 > 0.5 \end{cases}$$

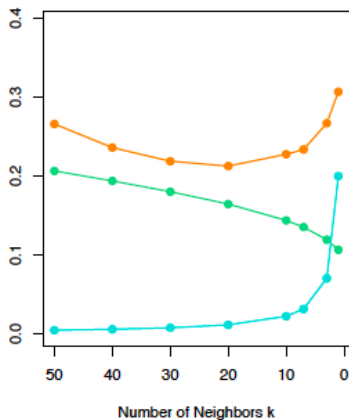
- ▶ Apply KNN to perform both the classification and regression tasks
- ▶ Use squared error loss to measure Err for the regression task
- ▶ Use 0-1 loss to measure Err for the classification task

Example 1

k-NN – Regression



k-NN – Classification



- ▶ Orange curve: expected prediction error
- ▶ Green curve: squared bias
- ▶ Blue curve: variance

Example 2

Set-up

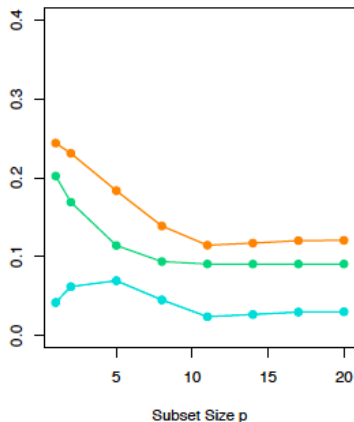
- ▶ $n = 80$ observations and $p = 20$ predictors
- ▶ X is uniformly distributed in $[0, 1]^{20}$ and

$$Y = \begin{cases} 1 & \text{if } \sum_{j=1}^{10} X_j \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

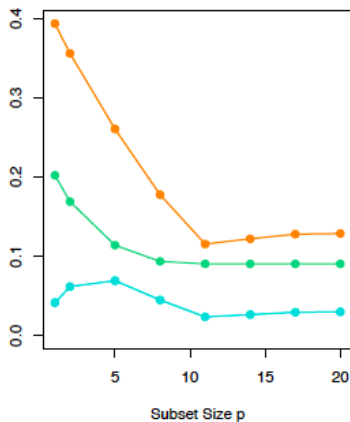
- ▶ Use best subset linear regression of size p for classification and regression tasks
- ▶ Use squared error loss to measure Err for the regression task
- ▶ Use 0-1 loss to measure Err for the classification task

Example 2

Linear Model – Regression



Linear Model – Classification



- ▶ Orange curve: expected prediction error
- ▶ Green curve: squared bias
- ▶ Blue curve: variance

Optimism of Training Error Rate

Optimism of $\overline{err}$

Suppose (x_0, y_0) is a new test data point, drawn from the joint distribution of the data.

$$\overline{err} = \frac{1}{N} \sum_{i=1}^N L(y_i, \hat{f}(x_i))$$

$$Err_T = E_{X_0, Y_0}(L(Y_0, \hat{f}(x_0)) \mid T)$$

Training error $\overline{err} \ll Err_T$ as it uses T for both fitting and assessment

Define in-sample error as

$$Err_{in} = \frac{1}{n} \sum_{i=1}^n E_{Y'}[L(y'_i, \hat{f}(x_i)) \mid T]$$

where the expectation is over new responses y'_i at each training point x_i

Optimism of $\overline{err}$

Define the optimism as

$$op = Err_{in} - \overline{err}$$

The average optimism

$$\omega = E_y[op]$$

where

- ▶ the training input vectors are held fixed
- ▶ the expectation is over the training output values

For squared error, 0-1, and other loss functions, one can show quite generally that

$$\omega = \frac{2}{n} \sum_{i=1}^n Cov(\hat{y}_i, y_i)$$

Optimism of $\overline{err}$

$$\omega = \frac{2}{n} \sum_{i=1}^n \text{Cov}(\hat{y}_i, y_i)$$

- ▶ The harder we fit the model, the stronger y_i and $\hat{y}_i$ are associated, and the larger ω
- ▶ Large $\omega \rightarrow$ greater optimism of $\overline{err}$

In summary, we have the important relation

$$E_y[Err_{in}] = E_y[\overline{err}] + \frac{2}{n} \sum_{i=1}^n \text{Cov}(\hat{y}_i, y_i)$$

How to Estimate Prediction Error

Option 1

- ▶ Estimate the optimism and add it to $\overline{err}$
- ▶ C_p , AIC , BIC work in this way for a special class of estimates that are linear in their parameters
- ▶ Can use in-sample error for model selection but not a good estimate of Err

Option 2

- ▶ Use cross-validation or bootstrap as direct estimates of the extra-sample Err

Estimates of Err_{in}

Err_{in} Estimate: C_p Statistic

If $\hat{y}_i$ is obtained by a linear fit with d inputs, then

$$\sum_{i=1}^n \text{Cov}(\hat{y}_i, y_i) = d\sigma_\epsilon^2$$

for the additive error model $Y = f(X) + \epsilon$

So (holds approximately when $N \rightarrow \infty$)

$$E_y[Err_{in}] = E_y[\overline{err}] + 2\frac{d}{n}\sigma_\epsilon^2$$

Adapting this expression leads to the C_p statistic

$$C_p = \overline{err} + 2\frac{d}{n}\hat{\sigma}_\epsilon^2$$

where $\hat{\sigma}_\epsilon^2$ is an estimate of the noise variance

Akaike Information Criterion

$$AIC = -\frac{2}{n} \loglik + 2\frac{d}{n}$$

where $\loglik = \sum_{i=1}^n \log P_{\hat{\theta}}(y_i)$, and $\hat{\theta}$ is the MLE of θ

- ▶ $-\frac{2}{n} \loglik$ rewards the fit between model and the data
- ▶ $2\frac{d}{n}$ is the penalty for including extra predictors in the model

AIC can be seen as an estimate of Err_{in} in this case with a log-likelihood loss

Akaike Information Criterion

Suppose we've got a set of candidate models $f_\alpha(x)$ indexed by α

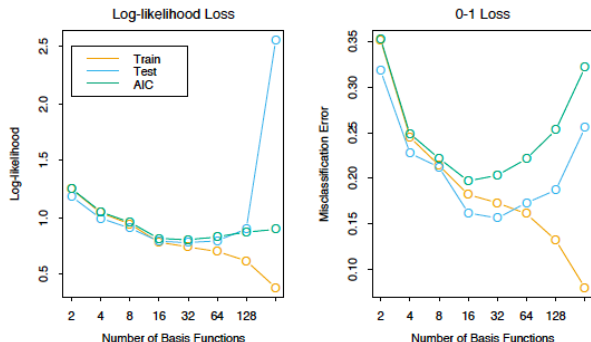
Denote $\overline{err}(\alpha)$ as the training error for corresponding candidate model

$d(\alpha)$ as the # of parameters for that model

$$AIC(\alpha) \cdot \hat{\sigma}_\epsilon^2 = \overline{err}(\alpha) + 2 \frac{d(\alpha)}{n} \hat{\sigma}_\epsilon^2$$

AIC can be seen as an estimate of Err_{in} in this case with a squared error loss

AIC for Model Selection



- ▶ Classifier is a logistic regression function with an expansion of M spline basis functions
- ▶ AIC is used to estimate Err_{in} with a log-likelihood loss
- ▶ AIC does well except when $M=256$ is large and $n=1000$
- ▶ Generally not recommended for 0-1 loss

Effective # of Paramaters

Effective Number of Parameters

For regularized fitting need to generalize the concept of number of parameters, why?

Consider regularized linear fitting - ridge, smoothing splines

$$\hat{y} = Sy$$

where

1. $y = (y_1, y_2, \dots, y_n)$ is the vector of training outputs
2. $\hat{y} = (\hat{y}_1, \dots, \hat{y}_n)$ is the vector of predictions
3. S is an $n \times n$ matrix, which depends on $x_1, \dots, x_n$ but not $y_1, \dots, y_n$

Define the effective number of parameters as

$$df(S) = \text{trace}(S)$$

Effective Number of Parameters

If S is an orthogonal-projection matrix onto a basis set spanned by M features, then $\text{trace}(S) = M$

$\text{tr}(S)$ - the correct quantity to replace d as the number of parameters in the C_p statistic.

If y arises from an additive error model $Y = f(X) + \epsilon$ with $\text{Var}(\epsilon) = \sigma_\epsilon^2$, then

$$\sum_{i=1}^n \text{Cov}(\hat{y}_i, y_i) = \text{trace}(S)\sigma_\epsilon^2$$

So the more general definition of dof is

$$dy(\hat{y}) = \frac{\sum_{i=1}^n \text{Cov}(\hat{y}_i, y_i)}{\sigma_\epsilon^2}$$

Bayesian Approach & BIC

Generic Form of BIC

Bayesian Information Criterion

$$BIC = -2\loglik + (\log N) \cdot d$$

- ▶ BIC statistic ($\times \frac{1}{2}$) is also known as the Schwarz criterion (SBC)

Under Gaussian model with known variance σ_ϵ^2 , for squared error loss

$$-2\loglik \propto \sum_i \frac{(y_i - \hat{f}(x_i))^2}{\sigma_\epsilon^2} = \frac{N \cdot \overline{err}}{\sigma_\epsilon^2}$$

As a result, BIC can be re-written as

$$BIC = \frac{n}{\sigma_\epsilon^2} \left(\overline{err} + \log(N) \frac{d}{N} \sigma_\epsilon^2 \right)$$

- ▶ BIC is proportional to AIC, with factor 2 replaced by $\log N$

Derivation of BIC

Starting point:

- ▶ $\{M_1, \dots, M_m\}$, a set of candidate models and their corresponding parameters $\theta_1, \dots, \theta_m$

Goal:

- ▶ Choose the best model M_i

How

- ▶ Training data $Z = \{(x_1, y_1), \dots, (x_n, y_n)\}$
- ▶ Priors $p(\theta_m \mid M_m)$
- ▶ Derive the Posterior of model M_m

$$\begin{aligned} P(M_m \mid Z) &\propto P(M_m)p(Z \mid M_m) \\ &\propto P(M_m) \int p(Z \mid \theta_m, M_m)p(\theta_m \mid M_m)d\theta_m \end{aligned}$$

Derivation of BIC

In comparison of two models M_m and M_l

$$\frac{Pr(M_m|Z)}{Pr(M_l|Z)} = \frac{Pr(M_m)}{Pr(M_l)} \times \frac{Pr(Z|M_m)}{Pr(Z|M_l)}$$

- ▶ Posterior odds = prior odds * BF(Z)
- ▶ $BF(Z) = \frac{Pr(Z|M_m)}{Pr(Z|M_l)}$ is called Bayes factor
- ▶ Indicates the contribution of the data toward the posterior odds

Derivation of BIC

The posterior of model M_m is

$$P(M_m | Z) \propto P(M_m) \int p(Z | \theta_m, M_m) p(\theta_m | M_m) d\theta_m$$

- ▶ Usually assume uniform prior $P(M_m) = 1/M$
- ▶ Approximate $\int p(Z | \theta_m, M_m) p(\theta_m | M_m) d\theta_m$ by simplification and Laplace approximation

$$\log P(Z | M_m) = \log P(Z | \hat{\theta}_m, M_m) - \frac{d_m}{2} \log N + O(1)$$

- ▶ where $\hat{\theta}_m$ is MLE and d_m is # of free parameters in M_m

If we define the loss function as $-2 \log Pr(Z | \hat{\theta}_m, M_m)$, then it's equivalent to BIC

AIC vs BIC

If $M_{true} \in \{M_1, \dots, M_M\}$ then as $N \rightarrow \infty$

- ▶ BIC will select M_{true} with probability 1
- ▶ AIC will not, tends to choose models which are too complex

However, when n is small

- ▶ BIC often choose models which are too simple

In addition, for BIC

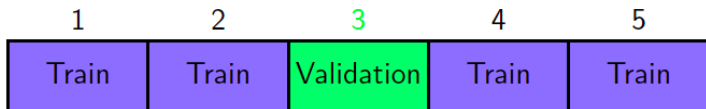
- ▶
$$\frac{e^{-\frac{1}{2}BIC_m}}{\sum_{l=1}^M e^{-\frac{1}{2}BIC_l}}$$
- ▶ Estimate not only the best model, but also their relative merits

Cross Validation

K-Fold Cross Validation

General Approach

- Split the data into K roughly equal-size parts



- For the k_{th} part, calculate the prediction error of the model fit using the other $K-1$ parts
- Do this for $k = 1, 2, \dots, K$ and combine the K estimates of the prediction error

When and Why

- It is applied when labelled training data is relatively sparse
- This method directly estimates $Err = E[L(Y, \hat{f}(X))]$

K-Fold Cross Validation

The mapping $\kappa : 1, \dots, n \rightarrow 1, \dots, K$ indicates observation i belongs to partition $\kappa(i)$

$\hat{f}^{-k}(x)$ is the function fitted with the k_{th} part of the data removed

Cross-validation estimate the prediction error

$$CV(\hat{f}) = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}^{-\kappa(i)}(x_i))$$

Typical choices for K are 5 or 10

The case $K = n$ is shown as leave-one-out cross-validation

K-Fold Cross Validation

Candidate models $f(x, \alpha)$ indexed by a parameter α

$\hat{f}^{-k}(x, \alpha)$ is the α_{th} model fit with k_{th} part of the data removed.

The define

$$CV(\hat{f}, \alpha) = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}^{-\kappa(i)}(x_i, \alpha))$$

Choose the model

$$\hat{\alpha} = \operatorname{argmin}_{\alpha} CV(\hat{f}, \alpha)$$

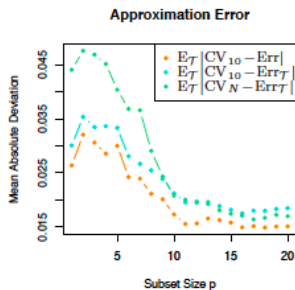
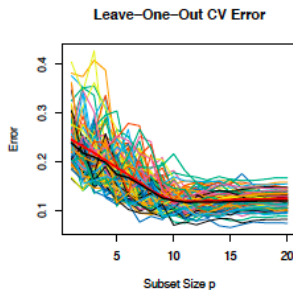
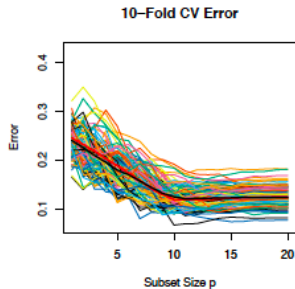
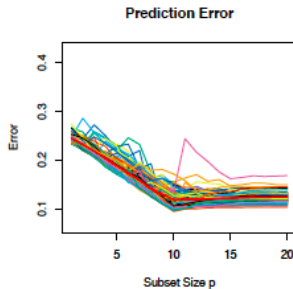
What Does Cross Validation Estimate

Intuition says

- ▶ When $K=5$ or 10 , $CV(\hat{f}) \approx Err$ as training sets for each fold are fairly different
- ▶ When $K = n$, $CV(\hat{f}) \approx Err_T$ as training sets for each fold are almost identical

However, according to theory and simulation experiments

- ▶ Cross validation really only effectively estimates Err



Thick red curve: Err ;

Thick black curve: $E_T[CV_K]$

What Values of K

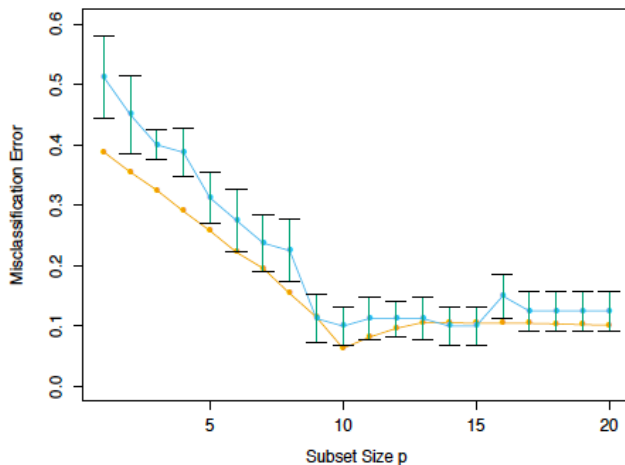
When $K = n$

- ▶ $CV(\hat{f})$ is approximately an unbiased estimate of Err
- ▶ $CV(\hat{f})$ has high variance as the n training sets are similar
- ▶ Computational burden high (except for a few exceptions)

When $K = 5$ (lowish)

- ▶ $CV(\hat{f})$ has low variance
- ▶ $CV(\hat{f})$ is potentially an upward biased estimate of Err

K-Fold Cross Validation Curve



Orange curve: Err_T ; Blue curve: $CV_{10}(\hat{f})$

Cross Validation - Right & Wrong

Task: classification problem with a large number of predictors

- ▶ $N = 50$ samples in two equal-sized classes
- ▶ $p = 5000$ quantitative predictors (standard Gaussian) independent of class labels
- ▶ True test error rate of any classifier 50%

Strategy 1

1. Screen the predictors: Find a subset of good predictors that are correlated with the class labels
2. Build a classifier: based on the subset of good predictors
3. Perform cross-validation: estimate the unknown tuning parameters and Err of the final model

Cross Validation - Right & Wrong

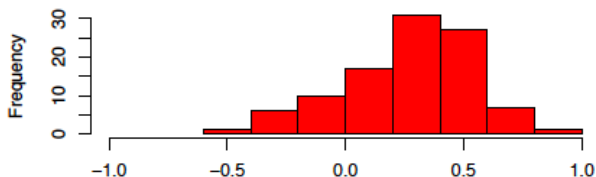
Task: classification problem with a large number of predictors

Strategy 2

1. Divide the samples into K groups randomly
2. For each fold $k = 1, \dots, K$
 - ▶ Find a subset of good predictors using all the samples minus the k_{th} fold
 - ▶ Build a classifier using all the samples minus the k_{th} fold
 - ▶ Use the classifier to predict the labels for the samples in the k_{th} fold
3. The error estimates from step 2 are then accumulated over all K folds to produce the cross-validation estimate of prediction error

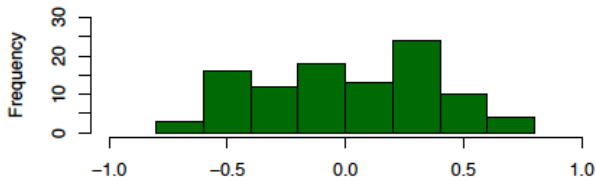
Cross Validation - Right & Wrong

Wrong way



Correlations of Selected Predictors with Outcome

Right way



Correlations of Selected Predictors with Outcome

Cross Validation - Right & Wrong

To perform multistep modeling

- ▶ Cross validation must be applied to the entire sequence of modeling steps
- ▶ Samples must be left out before any selection or filtering is applied which uses the labels

One exception - An unsupervised screening step can use all the samples

- ▶ select the 1000 predictors with highest variance across all 50 samples, before starting cross-validation
- ▶ this filtering does not involve the class labels, it does not give the predictors an unfair advantage

Bootstrap Method

The Bootstrap

Given a training set $X = (z_1, z_2, \dots, z_n)$ with each $z_i = (x_i, y_i)$

The **bootstrap** idea is

For $b = 1, 2, \dots, B$

1. Randomly draw n samples **with replacement** from Z to get Z^{*b} that is

$$Z^{*b} = (z_{b_1}, z_{b_2}, \dots, z_{b_n}) \text{ with } b_i \in 1, 2, \dots, n$$

2. Refit the model using Z^{*b} to get $S(Z^{*b})$

Examine the behavior of the B fits

$$S(Z^{*1}), S(Z^{*2}), \dots, S(Z^{*B})$$

Can Estimate Any Aspect of the Distribution of $S(Z)$

For example its variance

$$\hat{Var}[S(Z)] = \frac{1}{B-1} \sum_{b=1}^B (S(Z^{*b}) - \bar{S}^*)^2$$

where

$$\bar{S}^* = \frac{1}{B} \sum_{b=1}^B S(Z^{*b})$$

Use Bootstrap to Estimate Err

Attempt 1

$$\hat{Err}_{boot} = \frac{1}{B} \frac{1}{n} \sum_{b=1}^B \sum_{i=1}^n L(y_i, \hat{f}^{*b}(x_i))$$

where $\hat{f}^{*b}(x_i)$ is the predicted value at x_i using the model computed from Z^{*b}

Is this a good estimate or not? Why

- ▶ Overlap between the training and testing sets

How could we do better?

- ▶ Mimic cross validation

Use Bootstrap to Estimate Err

Attempt 2: leave-one-out bootstrap

$$\hat{Err}^{(1)} = \frac{1}{n} \sum_{i=1}^n \frac{1}{|C^{-i}|} \sum_{b \in C^{-i}} L(y_i, \hat{f}^{*b}(x_i))$$

where C^{-i} is the set of bootstrap samples b not containing observation i

- ▶ Either make B large enough so $|C^{-i}| > 0$ for all i or
- ▶ Omit observation i from testing if $|C^{-i}| = 0$

About Leave-one-out Bootstrap

Pros

1. avoids the overfitting problems of $\bar{Err}_{boot}$

Cons

1. Has the training-set-size bias of corss-validation
2. Probability that observation i in bootstrap sample Z^{*b}
$$P(i \in Z^{*b}) = 1 - \left(1 - \frac{1}{n}\right)^n \approx 1 - e^{-1} = 0.632$$
3. Therefore the average number of distinct observations in Z^{*b} is $0.632n$
4. $\bar{Err}^{(1)}$ bias similar to two fold cross validation (usually upward)

To Alleviate Bias

Attempt 3: The .632 estimator

$$\bar{Err}^{(.632)} = .368\overline{err} + .632\bar{Err}^{(1)}$$

- ▶ Compromise between the training error $\overline{err}$ and the leave-one-out bootstrap estimate
- ▶ Derivation not easy
- ▶ The constant .632 relates to $P(i \in Z^{*b})$

However, .632 estimator does not do well if predictor overfits

Estimate the Degree of Overfitting

No-information error rate

$$\hat{\gamma} = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n L(y_i, \hat{f}(x_j))$$

- ▶ Estimate of the error rate of $\hat{f}$ if inputs and outputs were independent
- ▶ Note the prediction rule $\hat{f}$, is evaluated on all possible combinations of targets y_i and predictors x_j

Estimate the Degree of Overfitting

Relative overfitting rate

$$\hat{R} = \frac{\bar{Err}^{(1)} - \overline{err}}{\hat{\gamma} - \overline{err}}$$

- ▶ $0 \leq \hat{R} \leq 1$
- ▶ $\hat{R} = 0 \rightarrow$ no overfitting
- ▶ $\hat{R} = 1 \rightarrow$ overfitting or no-information value $\hat{\gamma} - \overline{err}$

Use Bootstrap to Estimate Err

Attempt 4: The .632+ estimator

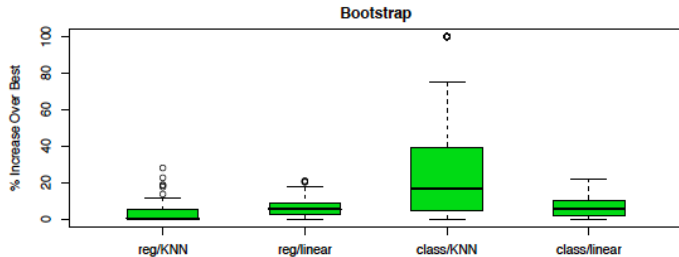
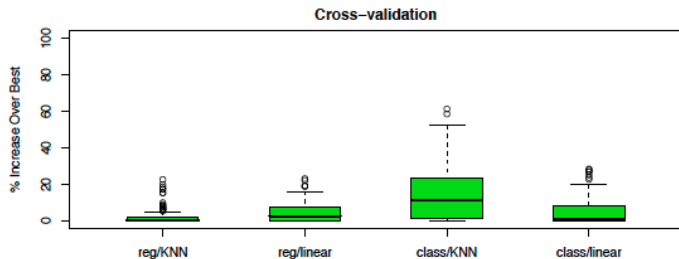
$$\hat{\text{Err}}^{(.632+)} = (1 - \hat{\omega})\overline{\text{err}} + \hat{\omega}\hat{\text{Err}}^{(1)}$$

with

$$\hat{\omega} = \frac{.632}{1 - .632\hat{R}}$$

- ▶ $.632 \leq \hat{\omega} \leq 1$ as $\hat{R}$ ranges from 0 to 1
- ▶ $\hat{\text{Err}}^{(.632+)}$ ranges from $\hat{\text{Err}}^{(.632)}$ to $\hat{\text{Err}}^{(1)}$
- ▶ $\hat{\text{Err}}^{(.632+)}$ is a compromise between $\hat{\text{Err}}^{(.632)}$ and $\overline{\text{err}}$ that depends on the amount of overfitting
- ▶ Again, the derivation is non-trivial

10-Fold Cross Validation & .632+ Bootstrap



$$\frac{Err_{\hat{\alpha}} - \min_{\alpha} Err(\alpha)}{\max_{\alpha} Err(\alpha) - \min_{\alpha} Err(\alpha)}$$

10-Fold Cross Validation & .632+ Bootstrap

In the above cases

- ▶ AIC, cross validation or bootstrap yields a model fairly close to the best available
- ▶ AIC (BIC) overestimate *Err* by 38%, 37%, 51%, and 30%
- ▶ Cross validation (bootstrap) overestimate *Err* by 1%, 4%, 0%, and 4%

However, for many adaptive, nonlinear techniques (like trees), estimation of the effective number of parameters is very difficult, making AIC impractical.